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SRN: PES1UG24CS043

1A

```
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$
```

1B

```
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$
```

2A

```
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$
```

2B

```
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ ./make_tree_dump
find .pes/objects -type f
tree object: cf664d59f2d705c7d6f011a27441fc025db0822befb3eea43d0fa7702bdb977b
.pes/objects/cf/664d59f2d705c7d6f011a27441fc025db0822befb3eea43d0fa7702bdb977b
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ ^C
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ xxd .pes/objects/cf/664d59f2d705c7d6f011a27441fc025db0822be
fb3eea43d0fa7702bdb977b | head -20
00000000: 7472 6565 2038 3700 3130 3036 3434 2061  tree 87.100644 a
00000010: 2e74 7874 0011 1111 1111 1111 1111 1111  .txt............
00000020: 1111 1111 1111 1111 1111 1111 1111 1111  ................
00000030: 1111 1111 1134 3030 3030 2073 7263 0022  .....40000 src."
00000040: 2222 2222 2222 2222 2222 2222 2222 2222  """""""""""""""""
00000050: 2222 2222 2222 2222 2222 2222 2222 22    """""""""""""""
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$
```

3A

```
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ ./pes add file1.txt file2.txt
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.c
  untracked:  Makefile
  untracked:  pes.h
  untracked:  tree.h
  untracked:  make_tree_dump
  untracked:  index.h
  untracked:  test_objects.c
  untracked:  tree.c
  untracked:  pes.c
  untracked:  commit.h
  untracked:  .gitignore
  untracked:  object.c
  untracked:  README.md
  untracked:  test_tree.c
  untracked:  test_sequence.sh
  untracked:  commit.c

akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$
```

3B

```
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ cat .pes/index
100664 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776671667 6 file1.txt
100664 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776671667 6 file2.txt
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$
```

4A

```
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ ./pes log
commit 10790146166fe85f6f6b7a858547c5a7814b9c49d692e0ecf897b3369a278962
Author: PES User <pes@localhost>
Date:   1776676426

    Add farewell

commit b6a9a26f7010293a1f84b29eadf669343695bd2f404b13399f2bad7fb6aab4e6
Author: PES User <pes@localhost>
Date:   1776676426

    Add world

commit 225f0ba47578b17af4eeafcb610c5d092669a3529b53dd53cf298ecd6edfca62
Author: PES User <pes@localhost>
Date:   1776676426

    Initial commit

akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$
```

4B

```
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/10/790146166fe85f6f6b7a858547c5a7814b9c49d692e0ecf897b3369a278962
.pes/objects/22/5f0ba47578b17af4eeafcb610c5d092669a3529b53dd53cf298ecd6edfca62
.pes/objects/50/da219c4c66a82940f7a8555908504a9bb1ede3773efe8669ad72f1d54966b3
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/7c/83d6df2d741034233b181c4bfa7a6e64a49ec875cc62a850886d11af72429c
.pes/objects/b6/a9a26f7010293a1f84b29eadf669343695bd2f404b13399f2bad7fb6aab4e6
.pes/objects/c1/5daab6ccf47288c923fa5384e07d0e7c63a4f443a1a77c77d51e67b257e822
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$
```

4C

```
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ cat .pes/refs/heads/main
10790146166fe85f6f6b7a858547c5a7814b9c49d692e0ecf897b3369a278962
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$
```

Integration Test

```
akshath@akshath-Alienware-m15-R4:~/Music/version_control/PES1UG24CS043-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 3296e992dbc5... Initial commit

Log after first commit:
commit 3296e992dbc5b4b7729fac5786afdbdb61664adac5a0c78427d08ec39af99edb
Author: PES User <pes@localhost>
Date:   1776676929

    Initial commit


--- Second Commit ---
Committed: 652cf6eb506c... Update file.txt

--- Third Commit ---
Committed: 618dbad222b3... Add farewell

--- Full History ---
commit 618dbad222b3f268493773cfbffcc8606448d9823a5fc8fb9b56c0c436cd2c69
Author: PES User <pes@localhost>
Date:   1776676929

    Add farewell

commit 652cf6eb506c6fcd6c110c6abd38f55b05e8d47fa99ed712570f92cdc88e449d
Author: PES User <pes@localhost>
Date:   1776676929
```

```
    Update file.txt

commit 3296e992dbc5b4b7729fac5786afdbdb61664adac5a0c78427d08ec39af99edb
Author: PES User <pes@localhost>
Date:   1776676929

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
618dbad222b3f268493773cfbffcc8606448d9823a5fc8fb9b56c0c436cd2c69

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/32/96e992dbc5b4b7729fac5786afdbdb61664adac5a0c78427d08ec39af99edb
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/61/8dbad222b3f268493773cfbffcc8606448d9823a5fc8fb9b56c0c436cd2c69
.pes/objects/65/2cf6eb506c6fcd6c110c6abd38f55b05e8d47fa99ed712570f92cdc88e449d
.pes/objects/84/c5fed72c2dd9a3130e2e9b98033827f62793fb80d0295ccedf5fc7972d96eb
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/de/34fb483a5c16ce29ec3195dd505dfda7e35cc5fe16a99da4b2455b332a93cf
.pes/objects/ec/0af37b159ae92a5850ceaa6e497860729d6158aab2c6833b6068a1e3c5e8ef
```

5.1

pes checkout <branch> updates .pes/HEAD to point to refs/heads/<branch> and reads the commit hash from .pes/refs/heads/<branch>. It then loads the commit's tree and updates the

working directory to match it: create needed files, delete extra ones, and restore contents from blobs. The index must also be updated to reflect the new tree.
This is complex because it must safely modify the working directory, handle nested trees, and avoid overwriting uncommitted user changes.

5.2
To detect conflicts, compare:

1. Working directory file vs index (hash comparison) → detects local modifications
2. Target branch version vs index → detects differences between branches

If a file is modified locally (working ≠ index) and also differs in the target branch (target ≠ index), checkout must refuse. This prevents overwriting uncommitted changes.

5.3
In detached HEAD, commits are created normally but are not attached to any branch. They form a chain but no branch points to them.
If the user switches branches, these commits may become unreachable. They can be recovered by creating a branch pointing to their commit hash before garbage collection deletes them.

6.1
Start from all branch heads and perform a traversal (DFS/BFS). Mark all reachable commits, trees, and blobs. Store visited hashes in a hash set.
After traversal, delete all objects not in the set.
For ~100k commits and 50 branches, only unique reachable objects are visited (not 50×100k), typically around ~100k plus trees/blobs.

6.2
GC is unsafe during commits because objects are created before references are updated.
Race condition: GC runs after objects are written but before the branch points to them → GC deletes them as “unreachable” → commit becomes corrupted.
Git avoids this using atomic writes, locks, and delaying deletion of recent objects.